# Lab1

1.

```java
public class Lab1{
    public static void main(String[] args) {
        System.out.println("Hello Visal!");
    }
}
```

```
Hello Visal!
PS D:\OOP>
```

2.

```java
import org.w3c.dom.Text;

public class Lab2{
    public static void main(String[] args){
        System.out.printf("%-10s  s\n", "\n", "Line break.");
        System.out.printf("%-10s  s\n", "\t", "Tabulation.");
        System.out.printf("%-10s  s\n", "\'", "Single quote.");
        System.out.printf("%-10s  s\n", "\"", "Double quote.");
        System.out.printf("%-10s  s\n", "\\", "\sign.");
        System.out.printf("%-10s  s\n", "\\\\", "\\Sign.");
        System.out.printf("%-10s  s\n", "//", "Line comment.");
        System.out.printf("%-10s  s\n", "/*...*/", "Block comment.");
        System.out.printf("%-10s  s\n", "\"\"\"\n\"\"\"", "Text block.");

    }

}
```

```
                    Line break.
                            Tabulation.
'                 Single quote.
"                 Double quote.
\                           ign.
\\                         \Sign.
//                Line comment.
/*...*/          Block comment.
"""
"""               Text block.
```

3.

```java
public class Lab3 {
    public static void main(String[] args) {
        System.out.println("************************");
        System.out.println(" **********************");
        System.out.println("  ********************");
        System.out.println("   ******************");
        System.out.println("    ****************");
        System.out.println("     **************");
        System.out.println("      ************");
        System.out.println("       **********");
        System.out.println("        ********");
        System.out.println("         ******");
        System.out.println("          ****");
        System.out.println("           **");
    }
}
```

```
************************
 **********************
  ********************
   ******************
    ****************
     **************
      ************
       **********
        ********
         ******
          ****
           **
```

3.1.

```java
public class Lab3One {
    public static void main(String[] args) {
       int rows = 10;
       int cols = 10;
       for(int i = 1; i <= rows; i++){
            for(int j = 1; j <= cols; j++){
                if(i == 1 || i == rows || j == 1 || j == cols){
                    System.out.print("* ");
                }
                else{
                    System.out.print("  ");
                }
            }
            System.out.println();
       }
    }
}
```

```
* * * * * * * * * *
*                 *
*                 *
*                 *
*                 *
*                 *
*                 *
*                 *
*                 *
* * * * * * * * * *
```

3.2.

```java
public class Lab3Two {
    public static void main(String[] args){
        for(int i=1; i<=5; i++){
            int j=i;
            for( j=i; j < i+5; j++){
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

```
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
```

4.

```java
public class Lab4 {
    public static void main (String[] args){
       System.out.print("    |");
       for(int j=1; j<=10; j++){
        System.out.printf("M|",j);
       }
       System.out.println();
       System.out.println("-------------------------------------------------------");
       for(int i=1; i<=5; i++){
         System.out.printf("M|", i);
         for(int j=1; j<=10;j++){
            System.out.printf("M|",i*j);

         }
         System.out.println();
       }
    }
}
```

```
    |   1|   2|   3|   4|   5|   6|   7|   8|   9|  10|
-------------------------------------------------------
   1|   1|   2|   3|   4|   5|   6|   7|   8|   9|  10|
   2|   2|   4|   6|   8|  10|  12|  14|  16|  18|  20|
   3|   3|   6|   9|  12|  15|  18|  21|  24|  27|  30|
   4|   4|   8|  12|  16|  20|  24|  28|  32|  36|  40|
   5|   5|  10|  15|  20|  25|  30|  35|  40|  45|  50|
```

4.1.

```java
public class Lab4One {

    public static void main(String[] args) {
        System.out.println("+----------------+---------+-------------------+-----+");
        System.out.printf("| %-14s | %-7s | %-17s | %-3s |\n", "Country", "Name", "Profession", "Age");
        System.out.println("+----------------+---------+-------------------+-----+");

        System.out.printf("| %-14s | %-7s | %-17s | %-3d |\n", "Germany", "Michael", "Computer Engineer", 19);
        System.out.println("+----------------+---------+-------------------+-----+");

        System.out.printf("| %-14s | %-7s | %-17s | %-3d |\n", "England", "Robert", "Artist", 34);
        System.out.println("+----------------+---------+-------------------+-----+");

        System.out.printf("| %-14s | %-7s | %-17s | %-3d |\n", "United Kingdom", "Julia", "Designer", 42);
        System.out.println("+----------------+---------+-------------------+-----+");

        System.out.printf("| %-14s | %-7s | %-17s | %-3d |\n", "United States", "Jo", "Actor", 21);
        System.out.println("+----------------+---------+-------------------+-----+");
    }
}
```

```
+----------------+---------+-------------------+-----+
| Country        | Name    | Profession        | Age |
+----------------+---------+-------------------+-----+
| Germany        | Michael | Computer Engineer | 19  |
+----------------+---------+-------------------+-----+
| England        | Robert  | Artist            | 34  |
+----------------+---------+-------------------+-----+
| United Kingdom | Julia   | Designer          | 42  |
+----------------+---------+-------------------+-----+
| United States  | Jo      | Actor             | 21  |
+----------------+---------+-------------------+-----+
```

5.

A.

```java
String text = "I Love my Home Country";
        System.out.println("Text length is: "+ text.length());
```

```
Text length is: 22
PS D:\OOP>
```

B.

```java
String text1 = "I love my little country";
        System.out.println("Result: "+ text1.substring(7));
```

```
Result: my little country
PS D:\OOP>
```

C.

```java
String text2 ="Hi Students!";
        System.out.println("\"i\" is at index: "+ text2.indexOf("i"));
        System.out.println("\"u\" is at index: "+ text2.indexOf("u"));
        System.out.println("\"e\" is at index: "+ text2.indexOf("e"));
```

```
"i" is at index: 1
"u" is at index: 5
"e" is at index: 7
PS D:\OOP>
```

D.

```java
String text1 = "Hi Students!";
        String text2 = "Students";
        String text3 = "Hi Students!";
        System.out.println("Text1 is equal to Text2: "+ text1.equals(text2));
        System.out.println("Text1 is equal to Text3: "+text1.equals(text3));
```

```
Text1 is equal to Text2: false
Text1 is equal to Text3: true
```

E.

```java
String text1 = "Hi Students!";
         String text2 = "Students";
         String text3 = "Teacher";
         System.out.println("Text1 contains Text2: " + text1.contains(text2));
         System.out.println("Text1 contains Text3: " + text1.contains(text3));
```

```
Text1 contains Text2: true
Text1 contains Text3: false
PS D:\OOP>
```

6.

| Employee | | Object1 | | Object2 | |
|---|---|---|---|---|---|
| **Datatype** | **Variable Name** | **Variable** | **Value** | **Variable** | **Value** |
| int | employeeID | employeeID | 1001 | employeeID | 1002 |
| String | name | name | Dara | name | Sok |
| String | position | position | Manager | position | Developer |
| double | salary | salary | 1500 | salary | 1200 |
| int | age | age | 30 | age | 26 |
| String | department | department | IT | department | Software |

```java
class Employee {
    int employeeID;
    String name;
    String position;
    double salary;
    int age;
    String department;
}
```

6.1.

| Car | | Object1 | | Object2 | |
|---|---|---|---|---|---|
| **Datatype** | **Variable Name** | **Variable** | **Value** | **Variable** | **Value** |
| String | brand | brand | Toyota | brand | Honda |
| String | model | model | Corolla | model | Civic |
| int | year | year | 2020 | year | 2022 |
| String | color | color | White | color | Black |
| double | price | price | 25000 | price | 27000 |

```java
class Car {
    String brand;
    String model;
    int year;
    String color;
    double price;
}
```

6.2.

| BankAccount | | | Object1 | | | Object2 | |
|---|---|---|---|---|---|---|---|
| **Datatype** | **Variable Name** | | **Variable** | **Value** | | **Variable** | **Value** |
| int | accountID | | accountID | 101 | | accountID | 102 |
| String | ownerName | | ownerName | Dara | | ownerName | Sok |
| double | balance | | balance | 1500 | | balance | 2000 |
| String | bankName | | bankName | ABA | | bankName | ACLEDA |
| String | accountType | | accountType | Saving | | accountType | Current |

```
class BankAccount {
    int accountID;
    String ownerName;
    double balance;
    String bankName;
    String accountType;
}
```

6.3.

| Video | |
|---|---|
| **Datatype** | **Variable Name** |
| int | videoID |
| String | title |
| String | creator |
| int | duration |
| int | views |

| **Object1** | |
|---|---|
| **Variable** | **Value** |
| videoID | 1 |
| title | Java Tutorial |
| creator | Dara |
| duration | 15 |
| views | 1200 |

| **Object2** | |
|---|---|
| **Variable** | **Value** |
| videoID | 2 |
| title | OOP Basics |
| creator | Sok |
| duration | 10 |
| views | 900 |

```java
class Video {
    int videoID;
    String title;
    String creator;
    int duration;
    int views;
}
```